# A Methods and Analysis Summary of a Time-Course Expression Clustering for Mouse and Human (+control) Samples

## 1: Overview

Three sample sets (human, human control, and mouse) taken from the same source (see 2: Data Source) were processed through the same pipeline. The pipeline contained ortholog-based gene matching, sample-level quality control, DESeq2 normalisation, gene filtering, and Mfuzz gene clustering. Each stage has been described below, in the order they were carried out. The thresholds and criteria, along with the rationale used for said thresholds and criteria, are also explained.
The entire analysis was performed in R. The complete pipeline has four main scripts; two scripts for the full time-course and two scripts for the first two hours, plus supporting utilities (which are described in 7: Software and Implementation.)
(This project was conducted under the supervision of Dr. Kuljit Sandhu, assistant professor at IISER Mohali.)

## 2: Data Source

Raw RNA-sequencing count data for all three datasets originate from Barry et al. (2019)[i], made available under a Creative Commons Attribution License, with the original data deposited at GEO accession GSE129014. To enable direct comparison between species, human and mouse gene identifiers were mapped to one-to-one orthologs using the MGI ortholog reference table (HOM_ProteinCoding.rpt), which reports only unambiguous 1:1 mouse–human orthology relationships; genes with one-to-many or many-to-many relationships were excluded at this stage. Each raw count matrix (human, human control, mouse) was filtered to retain only genes with a confirmed ortholog, and mouse gene symbols were relabeled to their human ortholog for cross-species alignment. The three matrices were then further restricted to the intersection of genes present in all three, yielding a common set of 5,758 orthologous genes carried forward into all subsequent analysis.
Table 1 summarises the design of the three input datasets.

Table 1. Input datasets (all restricted to the 5,758 common orthologous genes).

| **Dataset** | **Samples** | **Time range (min)** | **Sampling interval** | **Samples at ≤ 120 min** |
|---|---|---|---|---|
| Human | 61 | 0–600 | 10 min | 13 |
| Mouse | 145 | 0–600 | 4 min, with six 8-min gaps ‡ | 30 ‡ |
| Control (human) | 61 | 0–600 | 10 min | 13 |

*‡ The mouse series has no sample at 60 min or at five later time points, so it is not perfectly regular at 4-minute spacing; the ≤ 120-minute window therefore contains 30 rather than 31 samples.*

## 3: Sample Quality Control

### 3.1: Time-Point Extraction

The column names in each of the raw data files followed differing naming conventions; however, by taking only the numerical values that occurred immediately after an underscore (that was present in

all the column names), it was found to correctly recover the time-point. This also applied to cases where there were numbers present in the column name that were details of the cell line, or when extra text followed the time-point tag.

## 3.2: Sample Normalisation

Raw Counts were normalised using DESeq2. Size factors were estimated (estimateSizeFactors) to correct for library-depth differences between samples. This was followed by variance-stabilising transformation (vst, blind = TRUE) to place all samples on a comparable scale.

## 3.3: Outlier Elimination

Four complementary diagnostic techniques were applied to each of the datasets, each ordered chronologically by timepoint. The methods were as follows:
i) a per-sample boxplot of VST values, assessing whether normalization had placed all samples on a consistent scale;
(ii) a plot of mean VST value against time, assessing global temporal trend and identifying local anomalies;
(iii) a sample-sample Pearson correlation heatmap, retained in chronological rather than clustered order so that correlation could be assessed as a function of time-distance; and
(iv) a per-sample boxplot of each sample's *correlation* to every other sample in the dataset, providing a direct view of the heatmap finding.
Samples were identified as technical outliers if they were showing markedly reduced correlation to the other samples. This was observed through the heatmap, as an *isolated* pale band spanning *across* the heat-map in both the row and the column, and also through the box-plot, if the box-plot median was clearly separated from the baseline. When detected, the observation was corroborated with at least one other diagnostic; usually the mean-VST-over-time plot. The cluster centroids were also used as a secondary diagnostic. Samples meeting this criterion in only a single diagnostic without independent corroboration were not excluded.

Table 2 lists all excluded samples and the supporting evidence.

Table 2: Excluded Samples (technical outliers) by dataset

| Dataset | Excluded Samples | Rationale/Primary Evidence |
|---|---|---|
| Human | NBplusNog_450, NBplusNog_530, NBplusNog_290, NBplusNog_370 | Per-sample correlation boxplot (median correlation 0.90–0.98 vs. dataset baseline of ~0.99); confirmed by sample-sample correlation heatmap |
| Mouse | X.m_152, X.m_468 | Correlation heatmap (isolated pale cross-band), mean-VST-over-time plot (local minima), and correlation boxplot, in agreement |
| Control (Human) | X.E8_410 | Correlation heatmap and correlation boxplot |

*A recurring, lower-amplitude periodic pattern (approximate 100–150-minute interval) was additionally observed in the mouse dataset beyond the two excluded samples; this was not resolved to individual outlier samples within the scope of this analysis and is noted as a limitation in Section 9.*
*Also note that, in the first-two-hour processing, it was noticed that X.m_0/X.m_4 in mouse also showed two sudden changes in expression. This is also noted in the first paper and is most probably significant biology information. Hence, it has been left as-is.*

# 4: Gene-Level Quality Control

As a gene-level check to be used alongside the sample-level check, the mean normalised expression of each gene was examined. This was done on a log-10 histogram, and it was intended that genes with more than 3σ (or 3 standard deviations away from the mean) on the log scale would be flagged. The expected case was that the samples would show a bimodal distribution; that is, the samples would show a large number of (essentially) silent genes, and a broad right-skewed population (or the tail of the graph). The bimodal distribution was expected as the sample was an *unfiltered* gene set; All three datasets exhibited the expected behaviour. Additionally, no genes exceeded the 3σ threshold, and hence, no genes were excluded at this stage.

# 5: Gene Filtering

Genes were retained for downstream analysis in two steps.
First, genes with a mean DESeq2-normalisation count *greater than* 1 were retained. This was to remove genes with negligible expression from the dataset.
Second, genes were ranked according to the coefficient of variation (or CV = standard deviation/mean), which was calculated upon the VST values. Only genes in the upper quartile (or upper 25%) were retained.
The CV threshold for retention was fixed at 25% after some trial and error. During the full time course analysis, the initial threshold was set at 0.4, which was found to be too high, removing genes present in every dataset. After this error, a histogram was used (frequency vs CV) and it was found that the actual 'threshold' was much lower. For the human and control(also human) datasets, the CV was found to be around 0.005-0.07, while for mouse it was closer to 0.009-0.15. To maintain a conservative threshold while still having a sufficient number of genes, the top 25% of all variable genes were taken. Since each dataset had different CVs, the top 25% of each dataset was taken. (The position of the threshold can be seen using the histogram plots, mentioned in Section 8: Images and Graphs.) To that end, the final CV cut offs are 0.0261 (human), 0.0370 (mouse) and 0.0262 (control) for the full time course, and 0.0279, 0.0399 and 0.0240 respectively for the first two hours. The position of the cut-off can be seen in the histograms.

# 6: Soft Clustering (via Mfuzz)

## 6.1: Fuzzifier and Candidate Cluster Counts

For each dataset, the fuzzifier m was calculated independently from the filtered and standardized VST matrix using Mfuzz's mestimate function. The seed for the clustering was set at 42 to ensure the clusters formed in separate runs did not change patterns.
Candidate cluster counts of c = 2, 3, 4, 6, 8, and 10 were judged by using two methods; a pairwise cluster overlap which linked any clusters with an overlap greater than 0.1, and a direct visual comparison of each cluster's profile by observing their centroids. Both were done using Mfuzz's in-

built functions. The second method was necessitated because observation of the soft clusters was visually insufficient to estimate cluster overlap.
The overlap threshold was initially set at 0.5. However, this threshold was essentially useless, as it severely under-reported genuine overlaps between clusters that were clearly visible in the centroid clusters.

## 6.2: Cluster Count Selection

In the full time-course, for human and mouse, it was observed that exceeding cluster count over 3 (c = 3) would fragment a single cluster into mutually over-lapping and ever increasing redundant clusters, while two well defined clusters stayed separated no matter what the value of c. A value of c = 3 was kept, and three clearly separated non-redundant cluster centroids were achieved in each case. For control, keeping c = 3 showed two clusters moving fairly closely to each other, and was reflected in a (comparatively) low median of membership (0.79) and only about 62% of the genes exceeded a membership of 0.7. Reducing c to 2 (or c = 2) collapsed this into a single cluster with a median membership of 0.91, with 81% of the genes exceeding 0.7. Hence, c = 2 was used as the final solution.
In the first two hours, human and control samples showed an overlap in every c tested. Mouse remained well separated no matter what the c count was, with the first overlap occurring at c = 8. A common value of c = 3 was chosen for each dataset, for the sake of consistency across datasets. The choices were supported by the final numbers (see table 5), but naturally they are not as confident as the full time course.

Table 3. Cluster-count selection summary.

| **Time course** | **Dataset** | **Well separated at** | **Overlap first seen at** | **Final c** |
|---|---|---|---|---|
| Full | Human | $c \leq 3$ | $c = 4$ (redundant clusters) | 3 |
| Full | Mouse | $c \leq 3$ | $c = 4$ (redundant clusters) | 3 |
| Full | Control | $c = 2$ | $c = 3$ | 2 |
| ≤ 120 min | Human | none | $c = 2$ (smallest tested) | 3 |
| ≤ 120 min | Mouse | $c = 2, 3, 4, 6$ | $c = 8$ | 3 |
| ≤ 120 min | Control | none | $c = 2$ (smallest tested) | 3 |

*Note. Candidate counts tested were c = 2, 3, 4, 6, 8 and 10. Full time-course entries are summarised from the selection narrative above; ≤ 120-minute entries come from the overlap evaluation.*

## 6.3: Final Clustering Outcomes

The final cluster assignments, per-cluster membership scores, and all three datasets for the full time course are summarised in Table 4. The full clusters are provided in the accompanying soft-clustering result files.

Table 4. Gene filtering and final clustering summary, by dataset for full time-course.

| Dataset | Samples | Genes, mean > | Genes, final | Clusters (c) | Median |
|---|---|---|---|---|---|

| | retained | 1 | (CV filter) | | membership |
|---|---|---|---|---|---|
| Human | 57 / 61 | 4,800 | 1,200 | 3 | 0.974 |
| Mouse | 143 / 145 | 4,504 | 1,126 | 3 | 1.000 |
| Control | 60 / 61 | 4,415 | 1,104 | 2 | 0.911 |

Table 5: Final cluster counts and gene assignment confidence, first two hours only, by dataset

| Dataset | Samples retained | Genes, final (CV filter) | Clusters (c) | Median membership | % genes > 0.7 |
|---|---|---|---|---|---|
| Human | 13 / 13 | 1,220 | 3 | 0.810 | 65.9% |
| Mouse | 30 / 31 | 1,107 | 3 | 0.999 | 94.6% |
| Control | 13 / 13 | 1,131 | 3 | 0.719† | 52.7% † |

*"Samples retained" gives the sample count after outlier exclusion relative to the original sample count. "Median membership" refers to each gene's membership score to its own assigned cluster.*
*† For the control, c = 3 was used for consistency with the other datasets, although the same solution was rejected in the full time course (Section 6.2); its membership statistics are correspondingly weaker (see Section 9). For the control, 1,229 genes exceeded the CV cut-off; of those, 1,131 genes remained once the the mean > 1 filter was applied.*

Median membership and the percentage of genes above 0.7 depend on the fuzzifier $m$ as well as on how well separated the clusters are. As $m$ approaches 1, assignments approach hard (0 or 1) memberships, whereas larger values give fuzzier assignments. Because $m$ is estimated from the number of genes and samples (Table 1, 4 and 5), it is close to 1 for the full time course (1.036–1.057) but higher for the ≤ 120-minute human and control sets (1.228 and 1.229), which have only 13 samples each. Membership values should therefore be compared across datasets and time courses with caution. They are most informative within a single dataset, for example when comparing candidate $c$ values, as was done for the control in Section 6.2.

## 7: Software and Implementation

All analysis was done in R with the help of DESeq2(normalisation and VST), Mfuzz(clustering), and Biobase(data handling).
The first script was to match the genes to orthologs. As mentioned above, this was done using the MGI ortholog reference table (HOM_ProteinCoding.rpt). The script itself contains an alternative source, which is the live Ensembl database.
The pipelines in both cases (full time course and first two hours time course) were split into two files; one to diagnose the data, and the other for the clustering code. Both versions of the files (the full time course and the 120 minute time-course) share the same basic structure and the same helper module.

## 8: Images and Graphs

Images and graphs can be found in the repository, within the relevant folders. They include a CV vs Frequency histogram, a VST boxplot, a VST correlation boxplot, cluster centroids for c = 2, 3, 4, and a PDF with cluster overlap plots for each sample, for both time-courses.

# 9: Limitations

A few limitations apply across both the full timecourse and the 120-minute subset.
Cluster-count selection, while backed by the overlap evaluation and membership statistics, still involved a degree of visual judgement in comparing candidate centroid trajectories (see 6.1). The reported counts are the smallest cluster count at which no further genuinely redundant clusters were identified given the diagnostics available. The mouse dataset also retains a lower-amplitude periodic correlation pattern beyond the samples excluded as clear outliers (3.3); this was not resolved to individual outlier samples within the scope of this analysis.
For the less-than-120-min section, sample sizes are far smaller than the full timecourse (the former has about 13-31, the latter has 57-145.) Its cluster-overlap and membership statistics carry less weight, particularly for human and control. Both showed cluster overlap at every candidate c evaluated (6.1), and whose final c = 3 solution was chosen for consistency across datasets rather than because the evaluation clearly supported it (6.2). Control's subset result in particular has weaker gene-assignment confidence than the equivalent full time-course solution that was itself rejected for being too redundant (6.3, Table 4).

# References

Barry C, Schmitz MT, Argus C, Bolin JM, Probasco MD, Leng N, Duffin BM, Steill J, Swanson S, McIntosh BE, Stewart R, Kendziorski C, Thomson JA, Bacher R (2019). Automated minute scale RNA-seq of pluripotent stem cell differentiation reveals early divergence of human and mouse gene expression kinetics. PLoS Computational Biology 15(12): e1007543. https://doi.org/10.1371/journal.pcbi.1007543

Raw data: Gene Expression Omnibus accession GSE129014.
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE129014

Original authors' analysis code: https://github.com/rhondabacher/RobotNeuralDiffPaper

Appendix:

Table 6 lists all files associated with this analysis, grouped by role: original input data, intermediate files produced by one pipeline stage and consumed by the next, analysis scripts, final result files, and this report itself. Diagnostic figures generated for review during the analysis (correlation heatmaps, boxplots, cluster overlap and centroid preview plots) are outputs of the scripts listed below rather than separate files, and are not itemized individually here.

Table 6: File manifest

| File / path | What it is |
|---|---|
| 1_data/matched/*_matched.csv (x3) | Ortholog- and gene-set-matched counts, one per dataset |
| 2_scripts_includes_legacy/ filter_orthologs.R | Filters raw counts down to genes with a confirmed 1:1 ortholog |
| 2_scripts_includes_legacy/ pipeline_utils.R | Shared helper -- output path/folder handling |
| 2_scripts_includes_legacy/ 01_diagnostics.R, 02_clustering.R | Master copies, full timecourse |
| 2_scripts_includes_legacy/ 01_diagnostics_120min.R, 02_clustering_120min.R | Master copies, ≤ 120-min subset |
| 2_scripts_includes_legacy/legacy/ (4 files) | Older, superseded pipeline versions, kept for reference |
| 3_full_time_analysis/scripts/ | Pinned copies of the scripts as actually run for the full timecourse |
| 3_full_time_analysis/diagnostic_results/ data_analysis/ {1_human,2_mouse,3_control}/ | CV histogram, mean-expression histogram, VST boxplot, correlation heatmap -- per dataset |
| 3_full_time_analysis/diagnostic_results/ cluster_analysis/ {1_human,2_mouse,3_control}/ | Centroid previews (c=2,3,4) and the overlap evaluation PDF -- per dataset |
| 3_full_time_analysis/results/ {1_human,2_mouse,3_control}/ | Final cluster plot + Mfuzz results CSV -- full timecourse (see Table 2). Control carries both the c = 2 solution (final, reported) and the superseded c = 3 solution, kept for comparison (filenames suffixed _c2 / _c3). |
| 4_first_2_hours_analysis/scripts/ 01_diagnostics_2hr.R, 02_clustering_2hr.R | Scripts for the ≤ 120-min subset (Section 7) |
| 4_first_2_hours_analysis/ diagnostic_results/ ... | Same breakdown as above, for the ≤ 120-min subset (Section 3.3, 6.1) |
| 4_first_2_hours_analysis/results/ | c = 3 final for all three datasets -- results/ populated (see Table 5 for gene-assignment confidence and the control caveat) |

| | |
|---|---|
| 4_first_2_hours_analysis/README.md | Notes specific to the ≤ 120-min subset |
| 5_docs/Methods_Report.odt | This write-up |
| README.md / LICENSE | Repo overview and licence (MIT) |

i Barry C, Schmitz MT, Argus C, Bolin JM, Probasco MD, Leng N, Duffin BM, Steill J, Swanson S, McIntosh BE, Stewart R, Kendziorski C, Thomson JA, Bacher R (2019). Automated minute scale RNA-seq of pluripotent stem cell differentiation reveals early divergence of human and mouse gene expression kinetics. PLoS Computational Biology 15(12): e1007543. https://doi.org/10.1371/journal.pcbi.1007543

Raw data: Gene Expression Omnibus accession GSE129014.
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE129014

Original authors' analysis code: https://github.com/rhondabacher/RobotNeuralDiffPaper